

Xiaopu Wang

State College, PA • (814) 280 7998 • xmw5221@psu.edu

EDUCATION

Pennsylvania State University

Ph.D. in Statistics

2024 – Present
University Park, PA

University of Science and Technology of China (USTC)

M.S. in Statistics

2021 – 2024
Hefei, China

Nankai University

B.S. in Mathematics

2017 – 2021
Tianjin, China

PUBLICATIONS

- **Wang, X.**, He, Z., Liu, C., and Li, R. (2026). *Beyond Heuristic Tuning: Power-Calibrated LLM Watermarking*. **International Conference on Machine Learning (ICML)**.
- *Jiang, Y., *Zhao, B., ***Wang, X.**, *Tang, B., et al. (*co-first) (2025). *UKB-MDRMF: A Multi-Disease Risk and Multimorbidity Framework Based on UK Biobank Data*. **Nature Communications**. doi:10.1038/s41467-025-58724-3.
- **Wang, X.**, Wang, X., Zhang, A., and Wen, C. *Copy Number Variation Detection Based on Constraint Least Squares*. **Statistics and Its Interface**. doi:10.4310/23-SII814.
- Shen, Y., Wang, J., . . . , **Wang, X.**, et al. (2025). *CATI: A Medical Context-Enhanced Framework for Diagnosis Code Assignment in the UK Biobank Study*. **Artificial Intelligence in Medicine**. doi:10.1016/j.artmed.2025.103136.

RESEARCH EXPERIENCE

Power-Calibrated LLM Watermarking

Pennsylvania State University

2024 – 2026
University Park, PA

- Derived closed-form relationships connecting logit-based watermark parameters to green-token rates, detection power, and KL distortion.
- Converted parameter selection into a constrained optimization problem and demonstrated stronger power–distortion trade-offs than heuristic baselines across four model families and three text domains.

Marked Hawkes Processes for Recurrent Events in EHRs

Pennsylvania State University; collaboration with UNC

2025 – Present
University Park, PA

- Developing marked temporal point-process models that integrate longitudinal EHR events to estimate recurrent disease risks and patient trajectories.
- Built cohort-construction, preprocessing, and evaluation pipelines for real-world analysis of MIMIC-IV data.

Multi-Disease Risk and Multimorbidity Modeling

University of Science and Technology of China

2021 – 2024
Hefei, China

- Built large-scale data cleaning, feature-engineering, disease-prediction, and survival-analysis pipelines for UK Biobank.
- Developed interpretable analyses and visualizations of disease risk factors, multimorbidity patterns, and temporal disease progression.

Copy Number Variation Detection

University of Science and Technology of China

2021 – 2024
Hefei, China

- Formulated copy number variation detection as a constrained least-squares problem and designed an alternating minimization algorithm.
- Validated the method on simulated and real genomic data, demonstrating robust change-point recovery under heavy-tailed noise.

TECHNICAL SKILLS

Programming: Python, R, C++, SQL, Linux, Mathematica

Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, CNNs, RNNs/LSTMs, Transformers, diffusion models

Statistics & Data: Hypothesis testing, survival analysis, recurrent-event modeling, causal inference, large-scale data processing, visualization

LLMs & NLP: LLM watermarking, prompt engineering, BERT, GPT-family models, retrieval-augmented generation

Biomedical Data: UK Biobank, MIMIC-IV, All of Us, clinical NLP, comorbidity analysis